onboarding

Service Description

Service ID: *"onboarding"*

Abstract
This document describes the `onboarding` service, version 1.0.0, produced by the Onboarding support system of Arrowhead 5.2.2.

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1 Overview

This document describes the `onboarding` service, which provides the Arrowhead onboarding procedure: a device holding only an approved bootstrap certificate obtains its onboarding, device and system certificates and the identities of its systems. It is produced by the Onboarding system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 2, we describe the abstract message operations provided by the service. In Section 3, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2].

1.1 Significant Prior Art

Follows Bicaku et al., Interacting with the Arrowhead Local Cloud: On-boarding Procedure (2018), and the v4.6.2 onboarding controller.

1.2 How This Service Is Meant to Be Used

Step 1: with the bootstrap certificate, `onboard-with-csr` (or `onboard-with-name`) returns an onboarding certificate (`on`, at most 24 h). Step 2: with it, `onboard-device` returns the device certificate (`de`) and registers the device. Step 3: with the device certificate, `onboard-system` returns a system certificate (`sy`) and identity password for each approved system. The systems then register and orchestrate as usual. `offboard-device` withdraws everything.

1.3 Important Delimitations and Dependencies

Only devices approved with `onboardingManagement` (by bootstrap-certificate fingerprint) are accepted, and only the onboarding and device certificates issued last, while the CertificateAuthority reports them GOOD. Onboarding holds no CA key; the CertificateAuthority signs. Prefer CSRs: without one Onboarding generates the key and returns it once.

2 Service Interface

This section describes the interfaces to the `onboarding` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 3.

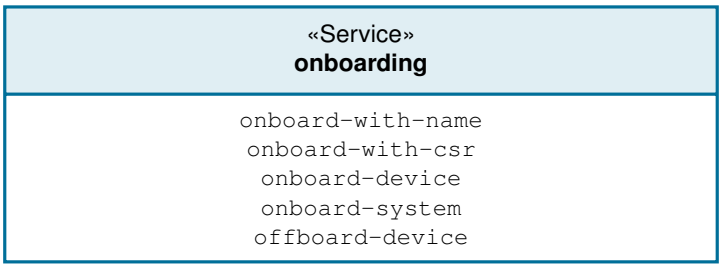


Figure 1: SysML-style block of the `onboarding` service and its operations.

The following interface operations are available.

2.1 operation onboard-with-name (NameRequest) : Response

Step 1 without CSR: Onboarding generates a key pair and returns the onboarding certificate with its private key (once).

2.2 operation onboard-with-csr (CsrRequest) : Response

Step 1: signs the device's onboarding certificate (`on`) from its CSR; returns the chain and the registry descriptors.

2.3 operation onboard-device (DeviceRequest) : DeviceResponse

Step 2, with the onboarding certificate: signs the device certificate (`de`) and registers the device in the ServiceRegistry.

2.4 operation onboard-system (SystemRequest) : SystemResponse

Step 3, with the device certificate: signs the system certificate (`sy`) of an approved system and creates its Authentication identity; returns the identity password once.

2.5 operation offboard-device () : OffboardingReport

With the device certificate: removes the device's systems, identities and registry entries, revokes the certificates and blacklists the systems.

3 Information Model

Here, all data objects that can be part of the `onboarding` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 3.16, which are used to represent things like hashes and identifiers.

3.1 struct Address

A network address with its type.

Field	Type	Description
type	String	Address type: `HOSTNAME`, `IPV4`, `IPV6` or `MAC`.
address	String	One network address (host name, IPv4, IPv6 or MAC address).

3.2 struct CertificateResult

An issued certificate.

Field	Type	Description
certificate	String	The certificate, Base64 DER.
certificateFormat	String	Encoding of the certificate, e.g. `X.509`.
certificateType	String	Certificate profile: `ON_BOARDING`, `DEVICE` or `SYSTEM`.
publicKey	String	Public key, Base64 X.509 SubjectPublicKey Info.
privateKey	String	Generated private key (only without a CSR, returned once).
expiresAt	DateTime	UTC expiry time; no value means no expiry.

3.3 struct CsrRequest

Onboarding with the device's own key.

Field	Type	Description
certificateSigning Request	String	PKCS\#10 certificate signing request, Base64 DER.

3.4 struct DeviceRequest

Request of the device certificate.

Field	Type	Description
certificateSigning Request	String	PKCS\#10 certificate signing request, Base64 DER.
addresses	List<String>	Network addresses of the device for the ServiceRegistry.
metadata	Metadata	Device metadata for the ServiceRegistry.

3.5 struct DeviceResponse

The device certificate and the next step.

Field	Type	Description
deviceName	Name	The onboarded device.
certificateChain	List<String>	Issued certificate first, then the issuing CA (Base64 DER).
deviceCertificate	CertificateResult	The issued device certificate (`de`).
approvedSystems	List<String>	Systems the operator approved for the device.
nextStep	String	The operation the caller performs next in the onboarding procedure.

3.6 struct DeviceResponseDTO

A registered device.

Field	Type	Description
name	Name	Device name.
metadata	Metadata	Free-form metadata (JSON object).
addresses	List<Address>	Network addresses with their types.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

3.7 struct NameRequest

Onboarding with a generated key for a name.

Field	Type	Description
commonName	String	Common name of the requested certificate, `<Name>.<Cloud>.<Organization>.arrowhead.eu`.

3.8 struct OffboardingReport

Outcome of an off-boarding.

Field	Type	Description
deviceName	Name	The off-boarded device.
completed	List<String>	Off-boarding steps that succeeded.
failed	List<String>	Off-boarding steps that failed.

3.9 struct Response

Result of the first onboarding step.

Field	Type	Description
certificateChain	List<String>	Issued certificate first, then the issuing CA (Base64 DER).
onboardingCertificate	CertificateResult	The issued onboarding certificate (`on`).
services	List<ServiceInstanceResponse>	Service instances of the core services the caller needs next.
nextStep	String	The operation the caller performs next in the onboarding procedure.

3.10 struct ServiceDefinitionResponse

A service definition.

Field	Type	Description
name	Name	Service definition name.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

3.11 struct ServiceInstanceInterfaceResponse

One interface of a service instance.

Field	Type	Description
templateName	Name	Name of the interface template, snake_case (e.g. `generic_https`).
protocol	String	Protocol of the interface, e.g. `http`, `https`, `mqtt`, `mqtts`.
policy	String	Security policy of the interface: `NONE`, `CERT_AUTH` or one of the token policies.
properties	Map<String, Object>	Interface properties defined by the template, e.g. `accessAddresses`, `accessPort`, `basePath`, `operations`.

3.12 struct ServiceInstanceResponse

A registered service instance.

Field	Type	Description
instanceId	Identifier	Service instance identifier `<System>|<serviceDefinition>|<version>`, or policy instance identifier.
provider	SystemResponseDTO	The provider system.
serviceDefinition	ServiceDefinitionResponse	The service definition.
version	Version	Version of the service instance.
expiresAt	DateTime	UTC expiry time; no value means no expiry.
metadata	Metadata	Free-form metadata (JSON object).
interfaces	List<ServiceInstanceInterface Response>	Service interfaces offered by the instance.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

3.13 struct SystemRequest

Request of a system certificate.

Field	Type	Description
systemName	Name	System to onboard; must be approved for the device.
certificateSigning Request	String	PKCS\#10 certificate signing request, Base64 DER.

3.14 struct SystemResponse

The system certificate, identity and the next step.

Field	Type	Description
systemName	Name	The onboarded system.
deviceName	Name	Name of the device hosting the system.
certificateChain	List<String>	Issued certificate first, then the issuing CA (Base64 DER).
systemCertificate	CertificateResult	The issued system certificate (`sy`).
authenticationMethod	String	Authentication method of the identity, `PASSWORD`.
credentials	Map<String, String>	Identity password of the new Authentication identity (returned once).
services	List<ServiceInstanceResponse>	Service instances of the core services the caller needs next.
nextStep	String	The operation the caller performs next in the onboarding procedure.

3.15 struct SystemResponseDTO

A registered system.

Field	Type	Description
name	Name	System name.
metadata	Metadata	Free-form metadata (JSON object).
version	Version	Version of the system.
addresses	List<Address>	Network addresses with their types.
device	DeviceResponseDTO	The device hosting the system, if registered.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

3.16 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
DateTime	Pinpoints a specific moment in time (UTC).
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
List	An *array* of a known number of items, each having type A.
Map	A set of key-value pairs.
Metadata	A set of key-value pairs with arbitrary, possibly nested values.
Name	A string identifier that is intended to be both human and machine-readable.
Object	Any structured value.
String	A chain of characters.
Version	Specifies a version, semantic versioning `major.minor.patch`.

4 References

[1] J. Delsing, “Onboarding – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-support-java-spring/Onboarding/documentation/v5.2.2/Onboarding-v5.2.2-SysD.pdf.

[2] ——, “onboarding HTTP/TLS/JSON – Interface Design Description (Onboarding),” Eclipse Arrowhead, IDD 5.2.2, 2026, onboarding-v5.2.2-onboarding-HTTP-IDD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	